

Supplementary Materials for “X-Gen: Enhancing Radiology Report Generation via LLM-Driven Data Augmentation and Decoupled Training”

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1. Examples of Augmented Reports

Figure 1 shows an example report from the IU X-ray training set and its augmentations in three different styles. Based on the results of the expert evaluation, we exclude “*formal and detailed*” style reports from future model training — more details about style selection are in Section 4.1.

2. Visualizations of Generated Reports

Examples of generated reports for IU X-ray, Abdomen X-ray, and MIMIC-mini are shown in Figure 2, Figure 3, and Figure 4, respectively. The contents highlighted in blue are matched anatomical locations, while yellow means matched clinical findings, and red represents matched medical devices. Decoupled training can indeed facilitate the generation branch learning medical semantics.

Taking generated reports of MIMIC-mini as an example, all three models generate the phrase “patient is status post median sternotomy and coronary artery bypass grafting cabg”, despite no relevant information in the ground truth report. Since MIMIC-mini is a randomly sampled subset of the MIMIC-CXR, it is difficult to control its diversity. The model tends to capture the most frequent phrase as a pattern. This phenomenon indicates the importance of diverse training data. On the other hand, X-CvT2GPT2-2 generates more matched entities due to its stronger image encoder, which enhances its ability of image understanding.

3. Questionnaire Designs

We present our questionnaire designs in Table 1.

4. Analysis of Different Styles

During the training for all three datasets, we record the top- K most frequently used style embeddings and cycle through them during inference i.e., for each test image, we

generate K reports. We demonstrate the evaluation results of each style for IU X-ray, Abdomen X-ray, and MIMIC-mini in Table 2, Table 3, and Table 4, respectively.

We can observe that there are seven active styles in the IU X-ray, eight active styles in the Abdomen X-ray, and only three active styles in the MIMIC-mini. The lack of styles in MIMIC-mini is due to the random sampling, which we have no control of the diversity of the data. For all three datasets, different style embeddings yield various performances. This result suggests that our training strategy effectively prevents style collapse, ensuring that each learned embedding remains distinct. Consequently, it is not possible to determine which embeddings correspond to synthetic styles and which to original styles. Given that the test set contains only original styles, we generate reports for each style embedding and select the one that gives the best performances.

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IU Chest X-ray			
Ground-Truth	Professional and Concise	Formal and Concise	Formal and Detailed
Cardiomediastinal silhouette is within normal limits of size and appearance. The pulmonary vascularity is unremarkable. Lungs are expanded and clear of airspace disease. Negative for pneumothorax or pleural effusion. Limited bone evaluation reveals no acute abnormality. There is incompletely evaluated lumbar levoscoliosis	Normal cardiomediastinal silhouette and pulmonary vascularity. Lungs expanded, clear, and free of airspace disease, pneumothorax, or pleural effusion. No acute bone abnormalities detected. Lumbar levoscoliosis incompletely assessed.	The cardiomediastinal silhouette presents normal size and appearance, with unremarkable pulmonary vascularity. Lungs demonstrate proper expansion and no signs of airspace disease, pneumothorax, or pleural effusion. Limited bone evaluation shows no acute abnormalities, while lumbar levoscoliosis is not fully assessed.	The examination of the cardiomediastinal silhouette reveals that it is within the standard parameters in terms of size and appearance. Upon assessing the pulmonary vascularity, no abnormalities were detected. The lungs are found to be expanded and devoid of any airspace disease. Additionally, there is no evidence of pneumothorax or pleural effusion. A limited evaluation of the bones indicates no acute abnormalities. However, it should be noted that there is an incompletely evaluated lumbar levoscoliosis present.

Figure 1: An example report from the IU X-ray training set and its augmentations in three different styles.

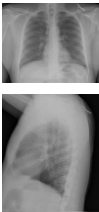
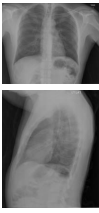
IU X-ray	Ground-Truth	Baseline-R2Gen	X-R2Gen-2	X-CvT2GPT2-2
	The heart and mediastinum are unremarkable. The lungs are clear without infiltrate . There is no effusion or pneumothorax .	The cardiomediastinal silhouette and pulmonary vasculature are within normal limits in size. The lungs are clear of focal airspace disease pneumothorax or pleural effusion . There are no acute bony findings.	The heart size and mediastinum are within normal limits. The lungs are clear. There is no focal airspace disease. No pleural effusion or pneumothorax . There is no acute bony abnormalities.	The heart and mediastinum are normal. The lungs are clear bilaterally. No evidence of infiltrate . No effusion or pneumothorax is present
	The heart is normal in size and contour . There is no mediastinal widening . The lungs are clear bilaterally . No large pleural effusion or pneumothorax . The XXXX are intact.	The cardiomediastinal silhouette and pulmonary vasculature are within normal limits in size. The lungs are clear of focal airspace disease pneumothorax or pleural effusion . There are no acute bony findings.	The heart size and mediastinal contours are within normal limits. The lungs are clear. There is no focal airspace disease. No pleural effusion or pneumothorax . There is no acute bony abnormalities.	The heart size and mediastinal contours are normal. No mediastinal widening is observed. The lungs are clear bilaterally . There is no significant pleural effusion or presence of pneumothorax .

Figure 2: Examples of generated reports for IU X-ray test set.

Abdomen X-ray	Ground-Truth	Baseline-R2Gen	X-R2Gen-2	X-CvT2GPT2-2
	The bowel gas pattern is within normal limits. No dilated loops of bowel to suggest obstruction . The visible visceral outlines are intact. No abnormal masses or calcifications . The visualized bony pelvis and spine is intact.	There are dilated loops of small bowel within the central and left upper quadrant. No air-fluid levels seen on erect projection. There are multiple air-fluid levels seen on the erect projection. No features of established bowel obstruction .	There is moderate faecal loading of the colon. The bowel gas pattern is within normal limits. There is no evidence of bowel obstruction . The visceral outlines are normal. No abnormal soft tissue mass or calcification seen. no acute focal bony abnormalities.	The bowel gas pattern is unremarkable. No evidence of dilated loops or bowel obstruction . The visceral outlines are clear. No abnormal masses or calcifications are observed. The pelvis and spine are normal.
	Some distension of loops of large bowel but absence of gas in the distal sigmoid and rectum . Appearances do raise some concern for distal large bowel obstruction developing. No free gas seen. Nasogastric tube position satisfactory.	No dilated loops of large or small bowel are identified to suggest established mechanical obstruction . Prominent gas-filled large bowel loop is seen in the left upper quadrant. No supine evidence of free intraperitoneal gas .	The abdominal x-ray examination consisting of supine and erect projections reveals a nasogastric tube with its tip located in the gastric body. Distension of loops of small bowel are observed in the left upper quadrant. There is an absence of free intraperitoneal gas . The nasogastric tube is in situ.	Supine and erect projections of the abdomen are provided. The bowel gas pattern in distal sigmoid and rectum is normal. Mild bowel obstruction is seen in the left upper quadrant. No evidence of free gas . The nasogastric tube is positioned correctly, and the stomach is distended..

Figure 3: Examples of generated reports for Abdomen X-ray test set.

MIMIC-mini	Ground-Truth	Baseline-R2Gen	X-R2Gen-2	X-CvT2GPT2-2
	There has been slight interval increase in size of bilateral pleural effusions which are now moderate to large. There is mild interstitial pulmonary edema , again noted. The heart size remains enlarged. The mediastinal contours are normal with note of calcification of the aortic knob . A left chest port catheter is stable in position. CHF , with slight interval increase in size of bilateral pleural effusions , now moderate to large.	The cardiac mediastinal and hilar contours are stable. There is mild to moderate pulmonary edema . Patchy opacities at the lung bases likely reflect atelectasis. no pleural effusion or pneumothorax is observed. no acute osseous abnormalities are detected.	There has been interval decrease in the right pleural effusion with associated compressive atelectasis at the left lung base. There is mild to moderate pulmonary edema . The aortic knob is tortuous with calcification . The left-sided ascapable and leads are unchanged in position with leads terminating in the right atrium and right ventricle. The cardiomeastinal silhouette is within normal limits.	Large bilateral pleural effusions is seen at the left lung base. There is mild pulmonary edema . The heart size and the mediastinal contours are within normal limits. The heart is stable in position. The aorta is tortuous with atherosclerotic calcifications at the aortic knob . There is evidence of Congestive Heart Failure . Retrocardiac opacity may be due to atelectasis though infection or aspiration cannot be excluded. Bony structures are intact.
	There is chronic blunting of the right lateral costophrenic angle , potentially due to scarring. The lungs are clear without consolidation or effusion . The cardiomeastinal silhouette is within normal limits. Median sternotomy wires and mediastinal clips are noted. Atherosclerotic calcifications noted within the tortuous thoracic aorta . No acute osseous abnormalities identified. No acute cardiopulmonary process .	The patient is status post median sternotomy and coronary artery bypass grafting cabg. The lungs are clear without focal consolidation or pleural effusion or pneumothorax. The cardiomeastinal silhouette is stable. No acute cardiopulmonary process.	Patient is status post median sternotomy and coronary artery bypass grafting cabg. The cardiomeastinal silhouette is mildly enlarged. Median sternotomy wires are again noted. pulmonary vasculature is not engorged. Lungs are hyperinflated without focal consolidation . No pleural effusion or pneumothorax is present. No acute osseous abnormalities are detected. No acute cardiopulmonary process is identified.	Patient is status post median sternotomy and coronary artery bypass grafting cabg. Blunting of the right costophrenic angle is noticed. Median sternotomy wires and mediastinal clips are intact and properly aligned. The aorta is calcified and tortuous . The lungs are clear without focal consolidation or pleural effusion . The cardiomeastinal silhouette is within normal limits. No acute osseous abnormalities are detected. No acute cardiopulmonary process is identified.

Figure 4: Examples of generated reports for MIMIC-mini test set.

Table 1: Overview of our questionnaire designs for expert evaluation.

(a) Headers of the questionnaire for Task 1. We omit the style tag during expert evaluation. Augment 1/2/3 refers to “*professional and concise*”, “*formal and concise*”, and “*formal and detailed*”, respectively. An example of evaluation is Y/N/Y, which indicates that “Augment 2” is not clinically accurate.

ID	Original	Augment 1	Augment 2	Augment 3	Doctor evaluation for augmented reports: Judge whether the augmented reports have the same semantics as the original report. (Y/N)
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(b) Headers of the questionnaire for Task 2. A “good” report is considered to be the style that resembles humans the most, while “bad” means the style is not practicable in real clinical settings.

ID	Report 1	Report 2	Report 3	Report 4	Doctor Evaluation: Ranking four reports regarding their writing styles from good to bad (1-4) <i>e.g.</i> , 2,3,1,4.
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Table 2: Evaluation results of different style embeddings for X-R2Gen-2 on the IU test set.

Dataset	Index	Bert-S	F1	RE-ME	RE-NLI	Rad-C	B-4	R-L	M	C
IU X-ray	0	0.611	0.490	0.455	0.501	0.331	0.189	0.409	0.212	0.409
	1	0.614	0.499	0.459	0.508	0.337	0.192	0.412	0.219	0.413
	2	0.609	0.487	0.452	0.499	0.329	0.184	0.405	0.219	0.396
	3	0.624	0.504	0.465	0.516	0.346	0.206	0.406	0.218	0.488
	4	0.618	0.500	0.460	0.506	0.340	0.198	0.409	0.202	0.389
	5	0.603	0.481	0.448	0.491	0.321	0.187	0.410	0.221	0.380
	6	0.595	0.477	0.439	0.482	0.315	0.186	0.388	0.218	0.340

Table 3: Evaluation results of different style embeddings for X-R2Gen-2 on the Abdomen X-ray test set.

Dataset	Index	B-1	B-2	B-3	B-4	R-L	M	C
abdomen X-Ray	0	0.394	0.245	0.162	0.111	0.310	0.149	0.252
	1	0.358	0.222	0.147	0.102	0.299	0.139	0.210
	2	0.423	0.263	0.174	0.119	0.312	0.156	0.279
	3	0.290	0.174	0.112	0.076	0.270	0.121	0.127
	4	0.439	0.273	0.180	0.122	0.316	0.160	0.292
	5	0.192	0.110	0.066	0.042	0.237	0.105	0.041
	6	0.322	0.200	0.133	0.092	0.288	0.130	0.157
	7	0.408	0.259	0.169	0.115	0.304	0.157	0.184

Table 4: Evaluation results of different style embeddings for X-R2Gen-2 on MIMIC-mini test set.

Dataset	Index	Bert-S	F1	RE-ME	RE-NLI	Rad-C	B-4	R-L	M	C
MIMIC-mini	0	0.491	0.401	0.336	0.209	0.185	0.094	0.271	0.133	0.138
	1	0.510	0.425	0.357	0.218	0.194	0.109	0.283	0.143	0.156
	2	0.508	0.422	0.354	0.215	0.190	0.107	0.280	0.140	0.153